A Comprehensive Approach to Transient Stability Control: Part II—Open Loop Emergency Control

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***Abstract*—The transient stability-constrained dispatch of an electric power system is a problematic task for the system operator who, under economic pressure, may be reluctant to take expensive preventive actions against very harmful contingencies. The open-loop emergency control (OLEC) technique proposed in this paper aims to relieve such preventive actions by complementing them with emergency ones. This is achieved by combining generation rescheduling, assessed and taken preventively, with generation tripping, assessed preventively but triggered only if the anticipated harmful contingency actually occurs. The relative size of preventive versus emergency control may be modulated so as to furnish panoply of solutions. Besides, the technique may stabilize simultaneously many contingencies. Simulations conducted on the EPRI 88-machine system illustrate various possibilities of the OLEC technique and compare it with the purely preventive one. Tradeoffs between preventive control and OLEC are also discussed, especially in the context of liberalized electricity markets. It is shown that OLEC is indeed able to provide a good compromise between economics and security, and to realize important savings.**

***Index Terms*—DSA, liberalized electricity markets, open loop emergency control, preventive control, SIME method, transient stability, transient stability control.**

I. Introduction

THE transient stability-constrained dispatch of an electric power system is a difficult task for the system operator, even in a vertically organized monopoly, where issues relating to individual interests do not arise. The task becomes much more problematic in liberalized electric industries, where market pressure makes the operator reluctant to take expensive preventive actions in order to guarantee stability, unless the system is in imminent danger of instability. A good compromise between economics and security may be achieved by complementing preventive control with emergency control.

The open-loop emergency control (OLEC) technique proposed in this paper precisely intends to mitigate the control actions taken preventively, by complementing them with emergency actions determined preventively but triggered in real-time only ***after*** the contingency occurrence. The preventive actions concern generation rescheduling, while the emergency actions concern generation tripping.

OLEC originates from the transient stability preventive control technique developed in the companion paper [1]. In short, this technique assesses instability in terms of margins and critical machines, and determines how much generation to shift from critical machines. Its combination with a standard OPF provides suggestions on how to redispatch this generation so as to meet additional objectives [1], [2].

Conceptually, OLEC may be considered as an extension of the purely preventive control. However, its countermeasures differ from preventive ones both in size and type. In particular, OLEC requires much less generation rescheduling; and the larger the amount of generation to trip correctively, the smaller the amount of generation to reschedule preventively. This provides a large combinatorial, out of which an "optimal" solution can be found.

From a practical point of view, the OLEC technique may be implemented in dedicated places, where generation tripping system protection schemes already exist. In such cases, OLEC could advantageously replace the traditional lookup tables derived from off-line studies (for a survey, see [3]) or even from on-line studies used to adapt the number of generators to trip to different operating conditions ([4] to [9]). Indeed, as already mentioned, OLECs innovation resides in a much faster and systematic on-line assessment of generation tripping and in its combination with generally inexpensive preventive generation shifting.

The purpose of this paper is to lay the foundations of OLEC, to highlight interesting features and possibilities, and to compare it with purely preventive control. Throughout, simulations carried out on the EPRI 88-machine system, modeled in the usual, detailed way, explore its performance and show its ability to relieve preventive countermeasures in a transparent way, which is especially interesting in the context of emerging liberalized electricity markets.

Note. [1], *which elaborates on SIME-based transient stability preventive control, is a prerequisite reading for the developments of this paper.*

II. Preventive versus Emergency Control

The general purpose of preventive control is to modify the operating conditions of a power system so as to make it able to withstand severe contingencies that would drive it to instability, should they occur. However, the preventive countermeasures advocated for very severe contingencies may be deemed too expensive, inasmuch as they are essentially concerned with active power. For example, simulations performed in [1] on the 88-machine EPRI system have shown that stabilizing ctg#1b requires shifting a very large amount of generation from the group of critical machines (over 50% of their total generation). The system

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operator usually would refuse to take such expensive preventive countermeasures for enhancing the system stability against contingencies that may not occur. An interesting alternative to such extreme solutions is emergency control. Here, the countermeasures are automatically triggered after a contingency has actually occurred and possibly cleared by appropriate protective devices.

Emergency control can broadly be classified into two categories: closed-loop and open loop [2], [10]. Closed-loop emergency control aims at assessing, ***on the basis of real-time measurements***, whether the contingency, which has actually occurred, is driving the system to instability; if so, at designing and triggering appropriate control actions and, further, at following-up the system evolution so as to make proper readjustments (additional control), if necessary. Although very appealing, the complete closed-loop cycle (predictive assessment of the instability and its size; design of corresponding control action; decision making and decision triggering) may be too slow to contain effectively the extremely fast developing transient instability phenomena (loss of synchronism may arise as quickly as, say, 150 ms). In such cases, open-loop control may be an interesting alternative. Indeed, its purpose is to trigger automatically the corrective action ***just after*** the contingency inception, by assessing the severity of the anticipated contingency ***on the basis of simulations***, and by arming the appropriate devices.

Today, transient stability emergency control relies mostly on open-loop schemes. Basically, these schemes are designed in a trial and error fashion, using transient stability simulations performed with standard time-domain programs [4] to [9]. A main difference with the technique proposed in this paper is that OLEC aims at realizing a tradeoff between preventive and open loop emergency control by ***combining preventive with emergency actions***. Besides, it uses a systematic assessment, able to provide various tradeoffs, out of which the operator may select an "optimum" solution. This optimum is at the crossroads of sufficiently moderate emergency control and economically acceptable preventive control. Section III lays the foundations of the OLEC technique.

Note. "Open Loop Emergency Control (OLEC)" refers to no feedback control. Admittedly, this term might suggest, "a man is into the loop" while, actually, "there is no man." "Feedforward emergency control" might be an interesting alternative. However, we prefer "OLEC," to emphasize the specific meaning of a mixture of predetermined preventive and emergency countermeasures.

III. OLEC TECHNIQUE

A. Principle

1) Description: The general procedure of OLEC is outlined below.

1) For an initially unstable scenario (operating condition subject to a predefined harmful contingency and its clearing scheme), compute the corresponding (negative) margin and determine the corresponding critical machines.
2) Assuming that (some of) these machines belong to a power plant equipped with a generation tripping scheme, select the number of units to trip in the emergency mode.
3) Run SIME, starting with the initial scenario up to reaching the assumed delay of generation tripping; at this time, shed the machines selected in Step 2, and pursue the simulation until reaching instability or stability conditions [see (1) and (3) of [1]]. If stability is met, stop; otherwise, determine the new stability margin and corresponding critical machines (to make sure that they have not changed from the previous simulation).
4) Run the transient stability control program (see [1, Sec. III.A]) to increase to zero this new (negative) margin. To this end, perform generation shifting in the usual way, from the remaining critical machines to noncritical machines. While performing this task, an additional objective may also be pursued, if the OPF software is combined with the transient stability control program [1], [11].
5) The new, secure operating state results from the combination of the above generation rescheduling taken preventively, and the consideration of the critical machines, previously chosen to trip correctively.
6) If there are more than one patterns of critical machines to trip, repeat the above Steps 1 to 5 with each one of them, until getting an operating condition as close as possible to the "optimum" one. The meaning of this "optimum" will be defined in Section III-B-II, and illustrated in Section IV-C.
7) After the "optimal" number of machines to trip is determined, the settings of the special protection activating the generation tripping scheme in the plant is adapted so as to automatically disconnect these machines in the event of the contingency occurrence.

2) Illustrative Example: In [1, Sec. III.C], contingency # 1b, has been stabilized in the preventive mode. Recall that the corresponding stability case has seven critical machines, belonging to the same power plant, and having a total generation of 5600 MW in the transient stability-unconstrained prefault operating condition. Recall also that stabilization of this contingency in the preventive mode requires decreasing this generation to 2423 MW (see [1, Table I]). Below, we consider again this contingency and stabilize it by OLEC, using the following parameters:

- number of critical machines (CMs) to trip: two; their stability-unconstrained prefault generation is: $821 + 769 = 1590$ MW (see in [1], column 1 of Table III);
- time delay for tripping these CMs: 150 ms.

Accordingly, the OLEC preventive generation rescheduling concerns the remaining five CMs, whose stability-unconstrained prefault generation is $5600 - 1590 = 4010$ MW.

The design of the control action follows the procedure of Section III-A-1. First, it is found that, because the case is very unstable, tripping 2 critical machines 150 ms after the fault inception would not be sufficient to stabilize it. Fig. 1 describes the dynamics of the stability case, respectively before [Fig. 1(a) and (b)] and after tripping the two critical machines [Fig. 1(c) and (d)]. More precisely, Fig. 1(a) and (b) portray, respectively,

TABLE I
STABILIZATION OF CTG # 1B BY OLEC (2 MACHINES TRIPPED)

1	2	3	4	5	6
It. #	η (rad/s)^2 (or MW)	Nr of CMs	Pc_k (MW)	Pc_{k+1} (MW)	sTDI (s)
0	-7.83*	5	4010	3609	0.421
1	-4.83*	5	3609	3260	0.461
2	-2.35*	5	3260	2662	0.506
3	-1.964	5	2662	2583	0.771
4	-1.265	5	2583	2440	0.856
5	-0.086	5	2440	2430	1.146
6	-0.036	5	2430	2415	1.216
7	0.028	5	**2415**	2422	5.000

Adapted from [11]

Recall that these simulations are performed for the same case with that displayed in Table I of [1], but considering that two machines are tripped 150 ms after the fault occurrence.

the multimachine swing curves and the δ-P OMIB curves of the totally stability-unconstrained system. Obviously, this is an extremely stressed system, for which the OMIB P_e and P_m curves do not intersect: there is no postfault equilibrium. Recall that in such cases the "standard" stability margin is replaced by the "substitute" one (minimum distance between these curves).

Fig. 1(c) and (d) portray the curves corresponding to the partly controlled case, obtained by tripping two critical machines 150 ms after the contingency inception. Comparing Fig. 1(a) and (c) suggests that the partly stability-constrained system is hardly less unstable than the unconstrained one: compare the slope of their swing curves as well as their times to instability (395 versus 421 ms). Similarly, comparing Fig. 1(b) and (d) shows that after the generation tripping (corresponding to $\delta_t = 7.34°$) the OMIB P_e curve gets somewhat closer to the P_m one, without, however, crossing it (their minimum "distance" decreases from 12.52 to 7.83 MW).

Since the emergency action is insufficient to stabilize the case, the procedure of Section III-A-1 continues with step 3: the new substitute margin is computed and the appropriate preventive generation rescheduling is decided in order to stabilize the operating conditions. The iterative procedure of rescheduling the five remaining machines is described in Table I. The asterisk in column 2 indicates minimum distances between P_e and P_m OMIB curves (in megawatt), to distinguish from "standard" margins expressed in $(\mathrm{rad/s})^2$. (See, for example, in row 0 of the table and in Fig. 1(d) the margin of -7.83 MW). Row 7, column 4 of Table I furnishes the final result of the preventive stabilization procedure of the system with five critical machines; it indicates that their generation should be reduced to 2415 MW.

Fig. 2 display the multimachine and OMIB δ-P curves corresponding to this stabilized case. They clearly show the effectiveness of the combined action of generation shifting (preventive) and generation tripping (emergency) controls. In particular, Fig. 2(b) shows that the OMIB equivalent power decreases from its initial value (-20.1 MW, see Fig. 1(d) to its secure value (-30.7 MW), creating a decelerating area and making the system able to dissipate the kinetic energy gained by the five critical machines in the during-fault period.

A synopsis of the seven simulations listed in Table I is given in Fig. 3, plotting the OMIB phase plane. Observe how clearly the system dynamics is described under the various conditions.

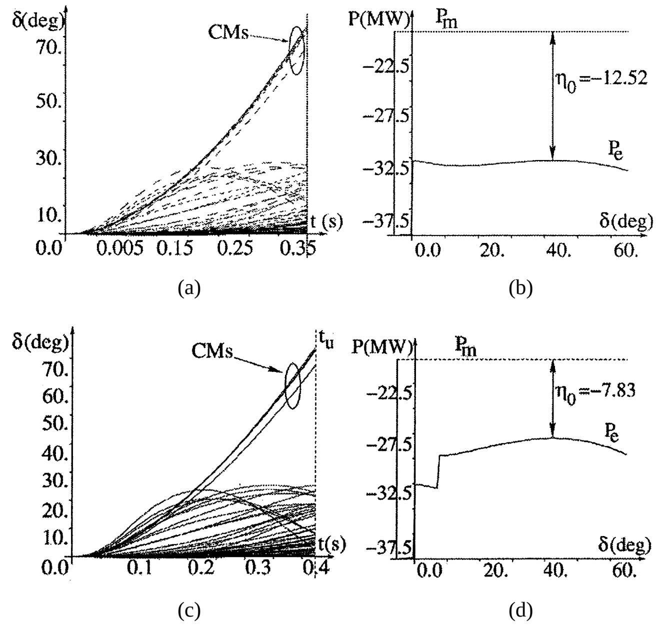

Fig. 1. Multimachine swing curves and δ-P OMIB curves of two unstable cases. (a) Swing curves, $t_u = 0.395$ s, (b) δ-P OMIB curves, (c) Swing curves, $t_u = 0.421$ s, (d) δ-P OMIB curves. Adapted from [11].

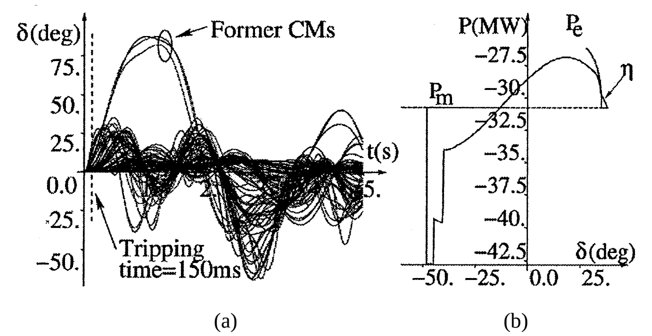

Fig. 2. Multimachine swing curves and δ-P OMIB curves of the stabilized case. (a) Multimachine swing curves, (b) δ-P OMIB Curves. $Pc_7 = 2415$ MW. Two generators are automatically tripped if the contingency actually occurs. Adapted from [11].

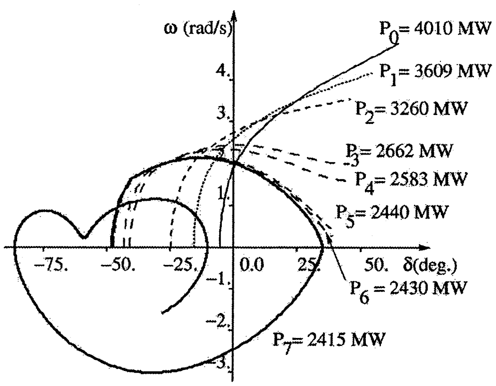

Fig. 3. OMIB phase plane representation of the seven simulations described in Table I. Two generators tripped if the contingency actually occurs. Adapted from [11].

In summary, to stabilize ctg # 1b by OLEC, the prefault generation of the five critical machines (CMs) should be decreased preventively from 4010 to 2415 MW, under the assumption that the other two CMs would be tripped correctively, after the contingency inception.

Once the conditions of the stabilization process are obtained, the special protection activating the generation tripping in the plant is armed to **automatically** disconnect the selected plants in case the contingency **actually occurs**. The remaining five machines are rescheduled to their new values. The prefault power of the plant, however, has been improved, with respect to the results presented in [1, Table I]. This is discussed below.

3) Comparisons and Discussion:

1) ***Purely preventive control versus OLEC***

Comparing the results of [1, Table I] and of this paper yields the following information.

- ***Stabilized prefault generation of the critical machines***

— preventive control: ***2423 MW*** (see [1, Table I])

— OLEC: $2415 + 1590 =$ ***4005 MW***

Thus, OLEC allows ***increasing the stable prefault generation*** of the plant from **2423 MW** (purely preventive control) to **4005 MW**. The gain is quite impressive.

Stated otherwise

- ***Total generation shift from the critical machines***

— preventive control: $5600 - 2423 = 3177$ ***MW***

— OLEC: $5600 - 4005 = 1595$ ***MW***.

2) ***Computing times of preventive control and OLEC***

Summing up the detailed sTDI of column 6 of Table I yields

— ***preventive control: 7.851*** sTDI

— ***OLEC: 10.377 sTDI***.

Thus, although larger, the computing time required by OLEC is still perfectly compatible with online computing requirements.

3) ***On the role of parameter*** t_u

The role of parameter t_u as a measure of the degree of instability is illustrated in column 6 of Table I where the sTDI values represent the corresponding "times to instability" (apart from the last one, which corresponds to a stable simulation, performed on the whole integration period).

B. Simultaneous Contingency Stabilization

1) Incentives: In preventive control, it is possible to stabilize simultaneously many harmful contingencies having common critical machines. This holds true also for OLEC, where, however, two main types of simultaneous stabilization should be distinguished. The one concerns cases with similar degrees of severity. In this case, the OLEC simultaneous contingency stabilization is an adaptation of the OLEC single contingency stabilization, just as in preventive control.

The other type of simultaneous stabilization concerns cases where some of the contingencies are significantly more severe than the others. In this case, the above adaptation could provide excessively expensive countermeasures. (For example, among the five harmful contingencies considered in [1], ctg#1b, which is, and by far, the most critical one, would impose significantly larger countermeasures than the other 4). An interesting alternative is the "hybrid stabilization" procedure described below.

2) Hybrid Simultaneous Contingency Stabilization: The leading idea is to decompose the considered harmful contingencies into two groups: the group of "relatively mild contingencies" (group 1) and the group of "significantly more severe contingencies" (group 2), and to proceed in two steps: the first stabilizes group 1 by preventive control, the second stabilizes group 2 by OLEC.

The resulting hybrid procedure is summarized below.

i) Stabilize simultaneously the contingencies of group 1, using preventive control, see [1, Sec. III.D]. Determine the resulting new generation of critical and noncritical machines.

ii) Stabilize the contingencies of group 2 using OLEC, starting with n machines to trip $(n = 1)$. The generation of these machines is fixed to the value determined in step (i), while the other machines' generation is modified according to OLEC.

iii) At the end of the OLEC stabilization, assess the stability of the contingencies of group 1. If at least one of them is found to be unstable go to step (i); otherwise, go to step (iv).

iv) If $n = n_{\max}$ stop. Otherwise, set $n = n + 1$ and go to step (ii). Here $n_{\max}$ denotes the admissible number of machines to trip.

The procedure is finished when the *"optimum" conditions* are reached. These are the conditions just necessary to stabilize contingencies of group 2 (the significantly more severe ones): tripping more machines would be needless and would overstabilize these contingencies.

The above procedure is illustrated below, in Section IV-C.

IV. Sample of Investigations

The first paragraph of this section illustrates the effect of generation tripping time on the secure prefault operating conditions. The last two paragraphs compare purely preventive control and OLEC, for single contingency and simultaneous contingency stabilization. The simulations are performed on the EPRI 88-machine system [14]. The various SIME-based software programs are coupled with the ETMSP transient stability program [15].

A. Prefault CMs Generation versus Triggering Time Delay

The time delay of the control action is an important parameter. It usually ranges in between 100 and 240 ms ([3], [5], [11]).

A series of simulations has been conducted, considering ctg #1b, when two critical machines are automatically tripped. The time delay is supposed to vary in between 0 (instantaneous activation of the scheme) and 400 ms [which is close to the time to instability, t_u, see Fig. 1(c)].

The numerical results, displayed in Fig. 4, corroborate common sense, namely, the earlier the emergency control action is applied the more effective the stabilization procedure.

TABLE II
SINGLE CONTINGENCY STABILIZATION. SECURE PREFAULT CRITICAL MACHINES' GENERATION VERSUS NR OF TRIPPED CMs

1 Case	2 Nr of TG *	3 TG Power	4 Nr of RG **	5 RG Power	6 Plant power	7 sTDI
0	0	0	7	2423	**2423**	8.265
1	1	820	6	2455	**3275**	8.175
2	2	1590	5	2415	**4005**	10.377
3	3	2408	4	2317	**4275**	4.456
4	4	3288	3	1923	**5151**	17.192
5	5	4010	2	1453	**5463**	13.043

*** TG=Tripped Generators. **RG= Remaining Generators.**

TABLE III
SIMULTANEOUS STABILIZATION OF ALL FIVE HARMFUL CONTINGENCIES. SECURE PREFAULT CMs' GENERATION VERSUS NR OF TRIPPED CMs

1 Case	2 Nr of TG *	3 TG Power	4 Nr of RG **	5 RG Power	6 Plant power	7 sTDI
0	0	0	7	2423	**2423**	26.825
1	1	705	6	2451	**3156**	48.090
2	2	1358	5	2439	**3797**	52.455
3	3	2063	4	2324	**4387**	50.902
4	4	2768	3	1960	**4728**	52.435
5	5	3438	2	1358	**4796**	44.750

*** TG=Tripped Generators. **RG= Remaining Generators.**

B. Single Contingency Stabilization

In this series of simulations, the secure prefault generation of critical machines is computed for a number of tripped generators varying in between 0 (corresponding to the pure preventive control) and 5. The triggering time delay is of 150 ms.

Table II shows that the total generation power of the critical machines of the prefault system condition P_{C0} varies in between 3275 MW (one critical machine supposed to be tripped) and 5463 MW (five critical machines to be tripped). Note that the value listed in row 0, column 6 of this table (2423 MW corresponding to the preventive mode stabilization) is the same with the one listed in Table I, row 5, column 4, of [1].

Comparing the values of column 6 of Table II with 2423 MW (critical machines' generation limit in the preventive mode) shows that OLEC allows increasing critical machines' generation from $3275 - 2423 = 852$ MW (one critical machine to trip) to $5463 - 242 = 3040$ MW (five critical machines to trip). Obviously, this corresponds to substantial savings: for example, decreasing the prefault total output of the plant by $5600 - 5151 = 449$ MW (four critical machines tripped if the contingency actually occurs) is much more acceptable than decreasing it by $5600 - 2423 = 3177$ MW in the pure preventive mode (for a contingency that may not occur).

C. Hybrid Simultaneous Contingency Stabilization

1) Description: The procedure of Section III-B-II is applied to the simultaneous stabilization of the five harmful contingencies considered in [1]. Here, group 1 (less harmful contingencies) is composed of contingencies # 30, 1a, 11, 10; group 2 (very harmful contingencies) is composed of ctg#1b only.

Step (i) of this procedure (simultaneous stabilization of contingencies # 30, 1a, 11, 10 in the preventive mode) has already been performed in [1, Table IV]. It was found that the total generation shifting required for stabilizing these four contingencies was 804 MW, distributed equally among the first seven critical machines that belong to the same power plant.

Hence, the power output of this plant was decreased from 5600 MW (initial prefault operating condition) to 4796 MW (stabilized condition).

The resulting generation rescheduling is then used to start stabilizing ctg#1b with increasing number of generators to trip (steps ii to iv of the procedure of Section III-B-2). Table III gathers detailed results. The tradeoff between critical machines' prefault generation decrease and number of critical machines to trip is also illustrated by the bar diagram of Fig. 5.

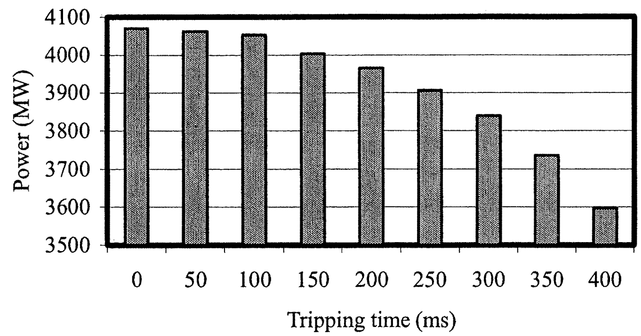

Fig. 4. Total stability-constrained power output of the generating plant versus OLEC's generation tripping time (five generators of the plant are rescheduled preventively and two generators are scheduled for tripping after the contingency inception).

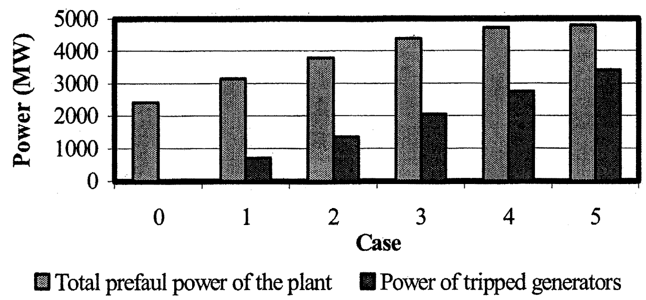

Fig. 5. Simultaneous stabilization of the five harmful contingencies. Secure prefault CMs' generation for various Nrs of tripped CMs.

2) Discussion and Comparisons:

1) Observe the important savings provided by the simultaneous OLEC stabilization, by comparing the value of row 0, column 6 of Table III with the other rows of the same column. (See also Fig. 5). For example, tripping curatively four machines (i.e., shedding 2768 MW; see column 3 of Table III) allows increasing the prefault total generation of the plant containing the initially seven critical machines to $4728 - 2423 = 2305$ MW.
2) Incidentally, note that the emergency mode stabilization requires a much larger amount of generation (to trip) than the shifted generation required for the preventive stabilization (3438 MW versus 2305 MW). This may be explained quite easily.
3) Strictly speaking, the ***"optimum number"*** of critical machines to trip using emergency control only would be five. Indeed, according to column 6 of Table III, in this

case, the prefault total generation of the plant amounts to 4796 MW (i.e., exactly the generation corresponding to the stabilization of the other four harmful contingencies, as just mentioned in Section IV-C-1). Notice, however, that OLEC makes it possible to refine the discrete emergency control action and trip four instead of five machines, with minor additional generation shifting from the remaining three machines (68 MW). This allows avoiding overstabilization and reducing by 670 MW the postfault imbalance between load and generation, if ctg # 1b occurs.

Of course, the final decision is left to the operator, who chooses the suitable control action taking into account the tradeoff between power shift and postfault imbalance.

4) Finally, column 7 of Table III provides computing times in terms of sTDI. Despite that they are much larger than (about twice as large as) those required for the simultaneous contingency stabilization in the purely preventive mode (26.825 sTDI, see row 0 of column 7), they are still compatible with real-time requirements.

V. Application of Near-Optimal Preventive Control & OLEC Techniques to Liberalized Electricity Markets

A. Online Security Assessment and Control

In the restructured electric industries, the operating condition of the system is set up in several cascading markets, which can be purely financial futures markets, used for expanding the system, or short-term markets for the physical delivery of electric energy. Generators scheduling is performed by bilateral contracts and auction markets that run one-day up to 1 h ahead of real time operation. After this new "unit commitment" process, the system operator conducts real-time markets in order to balance system generation and load and solve transmission constraints or "congestions." The specific rules of these real-time markets vary from one system to another, but they share a common feature: generators submit their prices for being redispatched up or down (if necessary) and the operator performs this security rescheduling with the economic objective of minimizing its total cost.

Most of the current markets take into account thermal constraints only. The near-optimal preventive control techniques proposed in [1] provide valuable tools for including transient stability limits in the auction mechanism of the balancing market. Note that these techniques may call upon an OPF software, as advocated in [1], or be adapted to any auction market software currently in use.

In the context of optional balancing markets, an additional advantage of the near-optimal preventive control techniques is that they can perform transient stability control using only a subset of the critical machines (see [1, Sec. III-C, III-D, and IV-D]). This allows meeting requirements regarding transparency of operator's decisions and nondiscriminatory access of all transactions to the transmission system.

Nevertheless, the examples described in the present paper show that, in some cases, the sole redispatch mechanism of the balancing market becomes very expensive. In such cases, the OLEC technique provides interesting alternatives, allowing considerable reduction of the power to redispatch preventively, and hence, of its cost.

Another interesting aspect of OLEC is that it can enhance significantly the security of the system by making economically possible its protection against the whole set of harmful contingencies, considered simultaneously.

OLEC is also helpful in adapting online the optimal number of generators to trip to the current system operating conditions, so as to minimize the imbalance between generation and load. Note that, depending on the amount of disconnected generation, this imbalance is generally compensated either by bringing on-line spinning reserve (in less severe cases) [3], and/or by shedding load (in the most constraining cases) [9][1].

A final notice: other congestion management methods, like counter trading [16], can also be adapted to include transient stability constraints, using SIME-based information about critical machines' identification and corresponding amount of power shifting.

B. Contracts for System Protection Schemes

Installing system protection schemes is significantly easier in vertical utilities than in liberalized electric industries where, however, such schemes could be fully justified. Indeed, experience has always shown that system protection devices have a positive impact on the overall financial account of the electric utilities, besides improving power system dynamic performance ([3], [13]). They allow operating the system under conditions very close to those established by economic considerations (minimum cost in traditional utilities, equilibrium of the auction market in restructured electric industries) while ensuring system stability, thus achieving a good compromise between economics and security.

In order to apply OLEC, the system operator can sign contracts with the companies owning the generating plants in which system protection schemes are installed. This type of contracts, which have already been implemented in some electricity markets [17], can also be used in other transient stability-constrained systems.

An advantage of using contracts for emergency control is that it allows mitigating considerably the preventive countermeasures, and hence, keeping almost unchanged the power system operating conditions established by the market; generally, the emergency control scheme remains inactive, since the triggering event does not occur often.

VI. Conclusions

A new transient stability OLEC technique has been proposed. It relies on the combination of preventive control with emergency control designed preventively but triggered only if the anticipated harmful contingency actually occurs. The ultimate objective is to relieve preventive countermeasures, so as to render them acceptable by the system operator, and, at the same time, to mitigate emergency control actions.

[1]Hence, whenever necessary, (i.e., if generation tripping is too important or generation rescheduling too difficult to realize), the amount of load to automatically shed in the emergency state should be assessed preventively by OLEC.

Specific procedures have been designed to stabilize the system against a set of harmful contingencies considered simultaneously. Of particular interest is the case of contingencies having significantly different degrees of severity.

Conceptually, OLEC preserves the advantages of SIME (flexibility with respect to power system modeling, contingency type and scenario, instability modes; robustness with respect to power system topology, operating conditions, contingency severity) together with advantages proper to control aspects (ability to assess the effect of realistic emergency control actions; ability to provide solutions with various types and size of emergency versus preventive countermeasures; ability to comply with online computing requirements; reliability of the advocated solutions, and their transparency).

The technique has been tested on realistic examples, which highlighted many of the above features. In particular, it was shown that OLEC may provide the operator with panoply of technical solutions, realizing significant savings in terms of generation rescheduling, and complying with various electricity market practices.

It was noticed that OLEC's implementation does not require special equipment, and may use existing dedicated sites while advantageously replacing existing software.

Admittedly, each real case has its specific requirements. This paper has concentrated on method's principle rather than an exhaustive consideration of such specifics. However, its great flexibility makes it easily adaptable.

Another issue, which has been mentioned but not addressed, concerns practices needed to avoid consequences of the imbalance caused by important generation tripping. Yet another issue worth to address is the combined use of transient stability open loop and closed-loop emergency control. Consideration of such issues is underway.

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